Progress report and action plan

Hardware Trojan detection dataset audit on **ver2** | Evidence through commit **261d326** | 3 November 2026

73	219	19 / 29	HOLD
verified host triplets	source design arms	original intervals now in band	training admission

Executive assessment

The current data are a **verified research cohort**, not an admitted training set. The expanded run produced 67 new verified triplets from 79 attempted candidates and combined them with six earlier independent-source triplets. Original, benign, and fault arms were kept within their host families. The frozen decision remains **HOLD**: 13 scalar or origin intervals miss the practical-equivalence band, 15 gate-type comparisons exceed the 5% limit, and eight typed comparisons are unresolved or unavailable. No pooled control reached the 0.90 hard-stop threshold, but that alone does not authorize training. [R1-R3]

The most actionable finding is a generator fingerprint. In the 17 fallback constructions, cell count ranks the fault above both negatives on **17/17 hosts**; in 50 native-edit families, the corresponding rankings are only **0.51** and **0.52**. The fallback procedure must be redesigned and the entire frozen check rerun. Three line-count origin controls also place the generated source outside the practical band relative to inspected TrustHub source. [R1, R2]

Recommended next move: fix construction and source-format asymmetries within existing verified lineages; verify ordinary and trigger behavior again; then repeat every frozen control before considering more host collection or detector training.

Scope of this report

This document consolidates the development pilots, the six-family overlap cohort, and the 79-family precision extension. It states what was measured, where the evidence is narrow, and the next admission gates. The archived baseline scores remain historical context, not evidence that the synthetic data improve Trojan detection. [R4-R8]

1 How the study reached this point

Stage	Result and lesson
Initial baseline and audit	A 42-circuit AES/RS232 fusion baseline recorded 41/42 correct (97.62%) under three-fold CV. A separate 10-sample PIC/memory-controller diagnostic was 50%. Metadata, lexical and host-family checks showed why a high development score could be misleading. [R4, R5]
Tier 1 / Tier 2	Nine control, storage, arithmetic and DSP hosts were built as matched variants. The first pilot failed a simple size shortcut (faults larger than originals on all nine). A second mixed-edit revision reduced that shortcut, but later typed-graph and independent-mutation checks found representation collisions and then a gate-histogram ranking of 27/27. [R6-R8]
Tier 3 and native TrustHub	A small RV32 pipeline and Wishbone interconnect plus AES/PIC native-host triplets added structural variety. Both new Tier 3 faults were smaller than benign edits; native variants differed from original controls. PIC mapped replay remained unknown. Source-origin and output-cone controls also failed. The admission protocol was frozen before this construction. [R9, R10]
Six external originals	Six pinned upstream families - divider, capture peripheral, SHA-2, AXI-Lite I2C, JPEG Huffman, PicoRV32 - passed the original-only normalized-token range and origin screen. All 18 arms passed bounded source/mapped checks; gate-type distances were at most 1.05%. Yet 29 pointwise intervals crossed the equivalence margin. [R3]
Precision extension	A plug-in width estimate called for as many as 69 total families, or 63 more. Sixty-three primary candidates and 16 supplementary candidates were locked before edits. Of 79 attempts, 67 new triplets verified. Reassessment separated insufficient precision from directional source and construction effects. [R1, R2, R11]

What remains comparable

All pilots are development evidence, but they are not one pooled benchmark. The nine Tier 1/2 hosts, the Tier 3/native hosts, the six independent-source hosts, and the TinyTapeout extension have distinct construction and selection histories. Family is the analysis unit. Variants or repeated submissions of a family never increase the effective host count. [R1, R3, R9]

2 Frozen admission design and evidence

Gate	Frozen requirement
Triplet	Unchanged original; verified equivalent benign edit; fault with a bounded, dormant-then-divergent output witness. Both negative comparisons are mandatory.
Build	Source and mapped simulation under the same Yosys flow; independently specified expectations where available; local SAT proof of benign expression.
Structure	Each fault-to-negative gate-type L1 distance / negative cell count at most 0.05; port/pin-aware typed comparisons must be distinguishable.
Shortcut controls	Source size/style, cell and edge counts, source-origin ranks against the inspected TrustHub panel, held-out learned metadata/graph probes, input fanout, output XOR and cone checks.
Decision	Hard stop for any tested paired score or origin AUROC at least 0.90 in either direction. Practical-equivalence pointwise 95% intervals must lie wholly within [0.30, 0.70]; at least six evaluation families and explicit motif coverage are required.

Expansion intake and verification

Originals came from pinned public TinyTapeout SkyWater submissions. The published chip netlist, with SkyWater functional models, served as an external oracle for a fixed reset, 16 idle cycles, then varied input bytes. Qualification required at least 90% fully defined oracle agreement, all normal cycles agreeing, and a nonconstant observation. Owner, author, title, top, file hashes, substantive module hashes and normalized token-shingle containment were checked to limit direct lineage duplication. The original RTL had to fall within the inspected TrustHub clean normalized-token range of 1,343-20,907. [R1, R3]

The generator first tried at most 28 native literal or input-bit edits per host. Benign arms retained the value; faults changed it. Seventeen hosts without a usable native witness received a separately tagged interface-wrapper fallback. Every accepted arm was replayed at source and mapped levels; the fault matched normal transactions and diverged on a recorded defined transaction. The fixed trace gives a bounded behavioral witness, not proof of arbitrary activation behavior or real-world stealth. [R1]

Verification outcome	Count
New originals screened and attempted	79
New verified triplets	67: 42 literal, 8 input-bit, 17 fallback
Unverified attempts retained in failure record	12: 11 synthesis/frontend problems; 1 mapped replay timeout
Combined behavioral cohort	73 families; 219 design arms
Complete learned graph-probe cohort	72 families; one verified fault maps to zero cells

3 What the expanded intervals actually say

The six-family pilot had 29 failed practical-equivalence intervals. All 29 pilot point estimates were inside [0.30, 0.70] and their intervals included 0.5. Using the observed host variation, the largest pointwise width projection was 69 total families; it was an estimate, not a power guarantee. After expansion to 73 verified families, **19 of those 29** intervals lie wholly inside the band; **seven** estimates lie inside but still cross a boundary; **three** line-origin estimates are outside the band and their intervals cross the boundary. All ten remaining intervals now exclude 0.5. This gives evidence of directional structure in the inspected cohort and rules out describing the entire problem as merely too few hosts. [R2, R11]

Type of remaining original failure	Examples (mean; 95% host CI)	Interpretation
Seven in-band estimates with wide intervals	Clean-vs-fault characters 0.623 [0.534, 0.712]; cells 0.630 [0.527, 0.726]; benign-vs-fault cells/edges 0.637 [0.548, 0.726]; fault-origin windows 0.350 [0.2999, 0.4025]	Margin unresolved; directional evidence exists.
Three out-of-band point estimates	Source-line origin: clean 0.288 [0.223, 0.356]; benign 0.288 [0.223, 0.356]; fault 0.264 [0.201, 0.330]	Likely source-format fingerprint; interval still crosses 0.30.
Newly failing origin controls	Source-origin cell count for each of the three arms has a lower CI below 0.30.	Thirteen total current interval failures, not just the original ten.

Interpreting the bootstrap

Intervals resample whole hosts and use exact percentile inversion for the saved score lattice. They are **pointwise**, conditional on the fixed TrustHub reference panel, selected families and held-out predictions. They are not simultaneous guarantees over all controls. The 73 designs share a TinyTapeout interface, technology and source-acquisition process; treating them as 73 fully independent samples from the full hardware population would overstate transfer evidence. Selection for token overlap also makes source-origin results development diagnostics rather than blind transfer estimates. [R1, R2]

4 Shortcut and representation audit

Check	Current observation	Consequence
Simple scalar and origin scores	No pooled score crosses the 0.90 hard stop; 13 equivalence intervals still fail. Maximum pooled class score reported: 0.681 (benign gate histogram); line-origin reaches roughly 0.736 in the opposite direction.	No single pooled trivial classifier completely solves the cohort; practical equivalence is still unproven.
Construction sensitivity	All 17 fallback faults have greater cell count than both negatives (17/17 for each comparison). Native-only 50-family rankings are 0.510 vs original and 0.520 vs benign.	Clear generator fingerprint; pooled scores hide a perfect subgroup shortcut.
Gate-type overlap	15 fault-to-negative comparisons exceed frozen 5% L1 limit.	Structural matching is incomplete; both negatives must pass per host.
Typed graphs	Eight comparisons remain inconclusive or unavailable, including the zero-cell fault; no graph feature is fabricated for it.	Some positive/negative distinctions cannot yet be established in the frozen representation.
Output motifs	Input fanout nonzero on 218 records; output XOR on 12; output cone on only 9.	Sparse output-motif coverage limits conclusions about these controls.

What the results do and do not establish

- Verification establishes bounded functional faults and benign equivalence at the selected sites. It does not establish malicious intent or realistic Trojan prevalence.

- The selected originals overlap the inspected TrustHub token-count range, but matching length does not make their RTL or gate topology representative of TrustHub.

- Strong within-host and origin controls are required before a detector result can be interpreted. A successful size gate alone would only close one shortcut.

- No new synthetic rows are admitted and no detector has been trained on these triplets. The earlier 97.62% baseline is a development result on different data.

5 Action plan with decision gates

Priority	Work	Exit evidence
1. Repair the generator	Retire or redesign the 17 interface-wrapper fallbacks so clean, benign and fault arms have comparable mapped footprints within each edit type. Try native internal edits or class-matched scaffolding with independently checked semantics. Keep the old failures visible; never silently filter them from a claimed final set.	A pilot where neither class can be ranked by construction type or simple cell/edge counts; both negatives preserved.
2. Correct source-format effects	Trace line-count and origin-cell differences to formatting, wrappers and synthesis choices. Apply one deterministic normalization to all arms and the TrustHub reference before feature extraction; retain original source for verification. Re-score raw and normalized views separately.	All mandatory source/style/origin intervals in band; no label-specific correction.
3. Strengthen structural checks	Resolve the zero-cell mapped fault and each inconclusive typed comparison, or exclude with a predeclared rule that keeps the failure record. Rework edits behind the 15 gate-type distances above 5%; rerun port/pin and cone controls, recording nonzero coverage.	Every retained host passes both gate-type comparisons and typed distinguishability; motif limits explicitly reported.
4. Reverify and freeze	For every revised triplet, rerun original-to-oracle, benign proof, normal trace, triggering witness and source/mapped agreement. Freeze generator, intake, scoring and thresholds before looking at new evaluation hosts.	Zero unverified retained records; hashes and complete failure manifest archived.
5. Broaden origin and holdout	Add new original host lineages from at least one other acquisition ecosystem, screen token overlap before editing, and keep one untouched evaluation panel. Group split by original lineage and report each ecosystem and construction type separately.	No trivial shortcut or origin rule passes a subgroup at hard-stop levels; sufficient eligible host families for meaningful intervals.
6. Train only after admission	If every frozen gate passes, train on an admitted development cohort; evaluate once on the untouched panel, with metadata-only, source-only, graph-only and motif controls. Report TrustHub and generated transfer in both directions separately.	Detector gain survives host-held-out and cross-generator comparisons; otherwise report the negative benchmark-validity finding.

Do not set a new family-count target from the old 69-family projection. The current blockers include systematic construction and source effects; more examples from the same generator could make their intervals narrower without making the data usable. A small repair pilot should be scored first, followed by a newly frozen, genuinely independent evaluation panel. [R1, R2]

Appendix A All 29 original interval outcomes

Expanded 73-family scores. The practical equivalence band is [0.30, 0.70]. 'Pass' means the complete pointwise interval is in band; 'Unresolved' crosses a margin with an in-band mean; 'Outside' has an out-of-band mean and crosses the margin. Source: [R2].

Domain	Arm	Control	Mean	95% CI	Status
class	clean	characters	0.623	[0.5342, 0.7123]	Unresolved
class	clean	comment_fraction	0.384	[0.2945, 0.4726]	Unresolved
class	clean	normalized_tokens	0.623	[0.5753, 0.6712]	Pass
class	clean	normalized_characters	0.623	[0.5342, 0.7123]	Unresolved
class	clean	cells	0.630	[0.5274, 0.7260]	Unresolved
class	clean	edges	0.596	[0.4932, 0.6918]	Pass
class	benign	characters	0.541	[0.5000, 0.5822]	Pass
class	benign	comment_fraction	0.459	[0.4178, 0.5000]	Pass
class	benign	normalized_tokens	0.500	[0.4795, 0.5205]	Pass
class	benign	normalized_characters	0.541	[0.5000, 0.5822]	Pass
class	benign	cells	0.637	[0.5479, 0.7260]	Unresolved
class	benign	edges	0.637	[0.5479, 0.7260]	Unresolved
origin	clean	characters	0.384	[0.3322, 0.4349]	Pass
origin	clean	lines	0.288	[0.2226, 0.3562]	Outside
origin	clean	windows	0.367	[0.3151, 0.4199]	Pass
origin	clean	comment_fraction	0.423	[0.3904, 0.4555]	Pass
origin	clean	normalized_windows	0.378	[0.3373, 0.4212]	Pass
origin	benign	characters	0.387	[0.3356, 0.4384]	Pass
origin	benign	lines	0.288	[0.2226, 0.3562]	Outside
origin	benign	windows	0.367	[0.3151, 0.4199]	Pass
origin	benign	comment_fraction	0.423	[0.3904, 0.4555]	Pass
origin	benign	normalized_windows	0.382	[0.3408, 0.4247]	Pass
origin	trojan	characters	0.366	[0.3165, 0.4160]	Pass
origin	trojan	lines	0.264	[0.2010, 0.3301]	Outside
origin	trojan	windows	0.350	[0.2999, 0.4025]	Unresolved
origin	trojan	comment_fraction	0.389	[0.3570, 0.4230]	Pass
origin	trojan	normalized_tokens	0.393	[0.3427, 0.4448]	Pass
origin	trojan	normalized_characters	0.370	[0.3202, 0.4217]	Pass
origin	trojan	normalized_windows	0.347	[0.3014, 0.3950]	Pass

Appendix B Sources, artifacts and limits

Repository evidence

[R1] experiments/precision_v1/README.md; data/precision_v1/admission.json; data/precision_v1/manifest.json; data/precision_v1/intake.json; data/precision_v1/reserve_intake.json.

[R2] data/precision_v1/reassessment.md; data/precision_v1/interval_review.md; data/precision_v1/planning.json.

[R3] experiments/overlap_six_v1/README.md; data/overlap_six_v1/admission.json; data/overlap_six_v1/overlap.json.

[R4] README.md, baseline and audit sections; retained baseline commit e061a06.

[R5] experiments/typed_fusion_v1/README.md; experiments/typed_fusion_v1/RESULTS.md.

[R6] experiments/tier12_v1/README.md; commit bb13402.

[R7] experiments/tier12_v2/README.md; commit b973d9b.

[R8] experiments/tier12_v3/README.md; commit 6a76933.

[R9] experiments/tier3_admission_v1/README.md; commit 0054107.

[R10] experiments/tier3_admission_v1/protocol.json; frozen at commit 7e7e7d0.

[R11] data/precision_v1/screen_provenance.json; data/precision_v1/lineage_check.json; data/integrated/ver2_precision_catalog.json.

Reproduction and status

The versioned source bundle is **data/precision_v1/triplet_edits.zip**; its SHA-256 is recorded in **data/integrated/ver2_precision_catalog.json**. The materializer **experiments/precision_v1/bundle.py --fetch** reconstructs all 201 newly verified source sets and checks hashes. Six earlier triplets reside in the separate overlap bundle. Raw run artifacts are in **runs/precision_v1**; the intake and failure manifests should remain with any report or reuse of the cohort. The current catalog explicitly records **status=hold**, **admitted_records=0** and **trained=false**. [R1, R3, R11]

Inference limits

All host bootstrap intervals are descriptive, pointwise and conditional on this inspected panel. Normal behavior and activation were tested on bounded traces. The original-only screen intentionally selected for token overlap and testability. The expanded cohort shares TinyTapeout interface conventions. The inspected TrustHub reference spans four families, and its source and mapped forms differ in provenance from the new submissions. These limits prevent a claim of deployment detection, benchmark interchangeability or independent evaluation performance. [R1-R3]